SUPPLEMENTARY

Item 1: Details of thiopurine-induced pancreatitis.

Study	Episodes of pancreatitis	Thiopurine-induced pancreatitis: details
Ardizzone 2003 (azathioprine versus methotrexate)	1	Pancreatitis was described as 'acute'. No information provided on timing, presentation, severity, clinical course of sequalae of pancreatitis. The patient was withdrawn from the trial.
Ardizzone 2004 (azathioprine versus mesalazine)	3	Pancreatitis was described as 'acute'. All three patients were withdrawn from the trial. No information provided on timing, presentation, severity, clinical course of sequalae of pancreatitis.
Cosnes 2013 (azathioprine vs placebo)	7	Adverse event described as 'serious'. No information provided on timing, presentation, clinical course or sequalae of pancreatitis. 6 of the patients with pancreatitis were switched to methotrexate and continued within the trial.
de Souza 2013 (azathioprine versus mesalazine)	1	Pancreatitis was described as 'acute'. No information provided on timing, presentation, severity, clinical course of sequalae of pancreatitis. The patient was withdrawn from the trial.
Ewe 1993 (azathioprine versus placebo)	1	Serum lipase concentration increased to 1002 U/L normal, < 190 U/L. This occurred in the 8th week of treatment. As the patient did not complain of any abdominal symptoms, treatment was continued. Lipase values returned to normal three weeks later and remained low.
Mantzaris 2009 (azathioprine vs budesonide)	2	No further details provided on timing, presentation, severity, clinical course or sequalae of pancreatitis. Both patients were withdrawn from the trial.
Mowat 2016 (6-mercaptopurine versus placebo)	1	Pancreatitis was described as 'moderate' in severity. No information provided on timing, presentation, severity, clinical course of sequalae of pancreatitis. The patient was not withdrawn from the trial.
Panes 2013 (azathioprine vs placebo)	7	All seven patients with acute pancreatitis were hospitalised. All patients diagnosed with acute pancreatitis were found to be symptomatic, had increased amylase and lipase levels, and had radiological evidence on cross-sectional imaging. No detail on timing, severity of sequalae of pancreatitis provided. All patients were withdrawn from the trial.
Present 1980 (6-mercaptopurine versus placebo)	1	No information provided on timing, presentation, severity, clinical course of sequalae of pancreatitis. The patient was withdrawn from the trial.
Reinisch 2010 (azathioprine versus mesalazine)	6	Pancreatitis was described as 'serious'. All six patients were withdrawn from the trial. No information provided on timing, presentation, severity, clinical course of sequalae of pancreatitis.
Savarino 2013	1	Participant developed severe abdominal pain at week 36

(azathioprine versus mesalazine)		with a concomitant increase in pancreatic enzymes. The patient was admitted to hospital and following withdrawal from azathioprine (and the trial), had complete relief of symptoms and normalisation of serum markers.
Summers 1979 (azathioprine vs placebo)	6	Details of the 6 episodes of pancreatitis are presented in a companion paper [15]. All episodes occurred between 13 - 21 days following dose 1 of azathioprine. Serum amylase / lipase ranged between 100 – 880 (% upper limit of normal for the laboratory). The time from dose 1 of azathioprine to amylase / lipase determination ranged from 17 – 35 days. Four of the patients were male, two were female. Five of the patients withdrew from the trial.

Item 2: Thiopurine-induced pancreatitis in leading Crohn's disease guideline publications

Guideline	**Reference to pancreatitis**	**Quantification of thiopurine-association pancreatitis risk**
British Society of Gastroenterology (BSG) consensus guidelines [21] Published June 2019	Recognises that thiopurines are contraindicated in pancreatitis, even at a low dose, due to high chances for recurrence. Does not state risk of occurrence on thiopurines.	No effect-size estimate provided.
National Institute for Health and Care Excellence (NICE) – NG129 [22] Published May 2019	Recognises that thiopurines are contraindicated if a patient has experienced past episodes of pancreatitis. Does not state risk of occurrence on thiopurines.	No effect-size estimate provided.
American College of Gastroenterology (ACG) clinical guideline [23] Published April 2018	Recognises that thiopurines are associated with pancreatitis.	No effect-size estimate provided.
Japanese Society of Gastroenterology (JSC) clinical practice guidelines [24] Published February 2018	Recognised pancreatitis as a side-effect of azathioprine and 6-mercaptopurine (evidence level: A)	No effect-size estimate provided.
European Crohn's and Colitis Organisation (ECCO) consensus guideline [25, 26] Published September 2016	Recognises that thiopurines are contraindicated if a patient has previous experienced episodes of pancreatitis. Does not state risk of occurrence on thiopurines.	No effect-size estimate provided.
World Gastroenterology Organisation (WGO) global guidelines [27] Published August 2015	No reference.	No effect-size estimate provided.
American Gastroenterological	No reference.	No effect-size estimate provided.

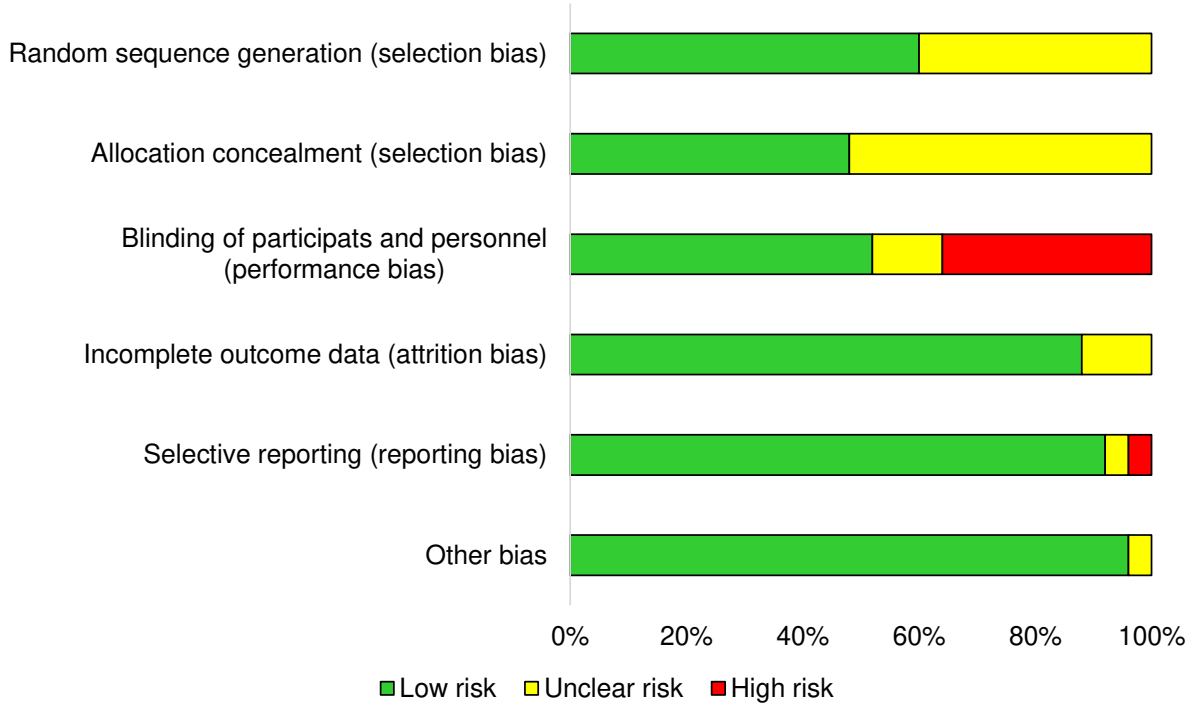
Association Institute guideline [28] Published December 2013		

Item 3: Methodological quality summary: review authors' judgements about each methodological quality item of included study.

	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Incomplete outcome data	Selective reporting	Other bias
Ardizzone 2003	+	?	-	+	+	+
Ardizzone 2004	+	?	-	?	?	+
Armuzzi 2013	?	?	-	+	+	+
Candy 1995	?	?	+	+	+	+
Colombel 2010	+	+	+	+	+	+
Cosnes 2013	+	+	-	+	+	?
D'Haens 2008	+	+	-	+	+	+
de Souza 2013	?	?	-	+	+	+
Ewe 1993	?	?	?	+	+	+
Hanauer 2004	+	+	+	+	+	+
Herfarth 2006	+	+	+	+	+	+
Klein 1974	?	?	+	+	+	+
Lémann 2005	?	+	+	+	+	+
López-Sanromán 2017	+	?	-	+	-	+
Mantzaris 2009	+	?	-	+	+	+
Markowitz 2000	+	+	?	+	+	+
Mowat 2016	+	+	+	+	+	+
O'Donoghue 1978	?	?	?	+	+	+

Panes 2013	+	+	+	+	+	+
Present 1980	+	?	+	+	+	+
Reinisch 2010	+	?	+	+	+	+
Rosenberg 1975	?	+	+	+	+	+
Savarino 2013	+	+	-	+	+	+
Summers 1979	?	+	+	?	+	+
Willoughby 1971	?	?	+	?	+	+

Item 4: Methodological quality graph: review authors' judgements about each methodological quality item presented as percentages across all included studies.


Random sequence generation (selection bias)
Allocation concealment (selection bias)
Blinding of participats and personnel (performance bias)
Incomplete outcome data (attrition bias)
Selective reporting (reporting bias)
Other bias
0%
20%
40%
60%
80%
100%
Low risk
Unclear risk
High risk